

DHRUV DUDEJA

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SUMMARY

Backend engineer with **2+ years** building production Python systems, specializing in **GenAI application infrastructure** — RAG pipelines, multi-model LLM orchestration, and concurrency-safe persistence layers. Delivered enterprise **ServiceNow** automation and knowledge ingestion systems in a regulated production environment. **AWS Certified Solutions Architect – Associate**.

TECHNICAL SKILLS

Languages: Python, SQL

AI & GenAI: LLMs (Claude Haiku 4.5, GPT-4o mini, GPT-5.4 mini), RAG Pipelines, Prompt Engineering, Multi-Model Orchestration, Vector Search, Metadata Schema Design, Recursive Summarization, Context Window Management, Structured Outputs

Frameworks & Tools: FastAPI, Mangum, PySpark, Pandas, Docker, Git, GitHub Actions, SAM

Cloud & Infrastructure: AWS (Lambda, API Gateway, Aurora Serverless v2, RDS, S3, Secrets Manager, Glue, EventBridge, SQS, KMS, CloudFormation), IaC, Serverless Architecture, Event-Driven Systems

Databases: PostgreSQL, Aurora PostgreSQL, Pinecone (Vector DB), Database Optimization (B-tree Indexing, CTEs, Table Partitioning, Materialized Views, Connection Pooling)

Practices: CI/CD, Concurrency-Safe Design, Idempotent APIs, Security Scanning (Wiz, Checkmarx), Observability, Agile

EXPERIENCE

Tata Consultancy Services | Software Engineer

Jun 2024 – Present

GenAI-Powered IT Change Request Automation Platform

Feb 2026 – Present

- **Built and own** the end-to-end Python backend for a production GenAI platform that automates enterprise IT Change Request (CR) generation in **ServiceNow** via a two-stage **RAG pipeline** (ticket summarizer → CR content generator), delivering **sub-10-second generation of complete 8-9 field structured CRs** and ~4-5s per-field-group regeneration.
- Designed **multi-model LLM orchestration** across **Claude Haiku 4.5** (CR generation), **GPT-4o mini** (ticket summarization and data-quality analysis), and **GPT-5.4 mini** (conversational refinement chatbot), with retrieval grounded on a **Pinecone** vector database of golden CRs and template-question corpora; scaled to **100+ change templates with architecture designed for 350+**.
- Engineered a custom **PostgreSQL-backed conversation persistence layer** replacing the internal LLM platform's native memory (which truncated history to 3-4 turns), enabling unbounded chat history via **recursive LLM-based summarization** with token-budgeted input/output caps and cumulative context compression across arbitrarily long sessions.
- Implemented **concurrency-safe writes** via a **two-phase-commit summarization protocol** (no DB lock across LLM calls), **epoch-CAS optimistic locking** with stale-lock recovery, and **idempotent inserts at message, session, and CR-submit layers** — preventing double-summarization, lost updates, and duplicate ServiceNow CR creation under concurrent-submit race conditions.
- Runs on **FastAPI + Mangum atop AWS Lambda** behind **API Gateway**, with **Aurora Serverless v2, S3, and Secrets Manager** provisioned via **CloudFormation IaC**; shipped through a **GitHub Actions CI/CD pipeline** with integrated **Wiz IaC and Checkmarx SAST security scanning**.

AI-Powered Incident Resolution & Routing Platform

May 2025 – Present

- **Built and own** the **Confluence knowledge ingestion pipeline** feeding a production **GenAI incident resolution system** on **ServiceNow** (RAG-based, **Pinecone**-indexed) — extracts pages and attachments via **Confluence REST API**, normalizes into a uniform embedding-ready format, and **designs metadata schemas stored alongside embeddings** to enable scoped and filtered vector search rather than blind top-k retrieval.
- Engineered the pipeline with **async processing + thread pooling, batched API calls** against ServiceNow and Confluence, **retry with exponential backoff**, and structured error logging — sustaining **99%+ availability** across recurring 15-day and monthly refresh cadences.
- **Designed and own** the backend for a production **event-driven serverless ticket routing system** on ServiceNow — **four AWS Lambda functions** across parallel INC and SR pipelines wired through **EventBridge cron + SQS with batch-failure retry semantics**, automatically assigning unassigned incidents and service requests to the right team member across enterprise support groups.
- Built the **candidate selection algorithm** powering both INC and SR pipelines — combining **shift-window filtering** including overnight UTC boundaries, **priority-weighted workload scoring** computed live from ServiceNow, and configurable per-group assignment rules — plus a **FastAPI admin backend** exposing team-config and audit endpoints, modeled across a **9-table Aurora PostgreSQL schema** deployed to dev/QA/prod via **SAM/CloudFormation** with GitHub Actions CI/CD and customer-managed KMS encryption.

Multi-Source Advertising Data Platform

Sep 2024 – Apr 2025

- Built a production **data ingestion pipeline** using **AWS Glue + PySpark** to ingest daily advertising data from **Meta (Facebook/Instagram), Snapchat, Twitter, Reddit, and Sprinklr** into a centralized **PostgreSQL** analytical store, exported to downstream teams via **FastAPI** endpoints.
- Optimized PostgreSQL via **CTE-based upserts, B-tree indexing, materialized views, table partitioning, and connection pooling** — reducing critical FastAPI response times **from 5-10 minutes to 4-5 seconds (~98% improvement)** by eliminating duplicate-row bloat in log-heavy tables lacking upsert semantics.
- Automated deployments via **GitHub Actions CI/CD** with **CloudFormation IaC** and replaced legacy **SFTP-based file transfer with Lambda-driven file copying** for core data workflows, delivering production hotfixes during off-hours to sustain **99%+ system uptime**.

PRIOR EXPERIENCE

Freelance Technical Writer — Produced technical documentation and instructional content. *Oct 2023 – Feb 2024*

Web Developer Intern, NullClass — Built responsive web templates and database structures. *Aug 2022 – Nov 2022*

CERTIFICATIONS

AWS Certified Solutions Architect – Associate

Oct 2025

Anthropic AI Fluency — Anthropic

2026

EDUCATION

B.Tech in Computer Science Engineering — Guru Gobind Singh Indraprastha University | CGPA: 8.7 *2019 – 2023*